

Jech 2

Vincent Wang

April 2025

1 Introduction

Suppose $f : A \rightarrow B$ is increasing and bijective. Then is $f^{-1} : B \rightarrow A$ increasing? Works if A is linear order. Let $b_1 < b_2$. Sps FTSOC $f^{-1}(b_1) \geq f^{-1}(b_2)$. Then $f(f^{-1}(b_1)) \geq f(f^{-1}(b_2))$, so $b_1 \geq b_2$, contradiction. Hence, $f^{-1}(b_1) < f^{-1}(b_2)$, so f^{-1} is order preserving. Statement: if A is a linear order and $f : A \rightarrow B$ is order preserving, then f^{-1} is also order preserving.

If $f : A \rightarrow B$ and $g : B \rightarrow C$ are order isomorphisms, then so is $g \circ f : A \rightarrow C$. $g \circ f$ and its inverse are bijective. Suppose $a < b$. Then $f(a) < f(b)$. Then $g(f(a)) < g(f(b))$. $g \circ f$ preserves order. Suppose $a'' < b''$ in C . Then $g^{-1}(a'') < g^{-1}(b'')$, so $f^{-1}(g^{-1}(a'')) < f^{-1}(g^{-1}(b''))$, so $(g \circ f)^{-1}$ is order preserving.

Let $f : W \rightarrow W$ be order preserving, W a well-ordered set. Then $f(x) \geq x$ for all $x \in W$. Suppose FTSOC there exists $f(x) < x$. Define nonempty $S = \{x \in W : f(x) < x\}$. Let m be the least element of S by Well-ordering. Then $m \leq s$ for all $s \in S$, so $f(m) \leq f(s) < s$ for all $s \in S$ as f is order preserving and by construction of S . $f(m)$ is a lower bound of S , so $m \leq f(m)$, a contradiction.

Proof. Let $S = \{x \in X : f(x) < x\}$ be nonempty and z be its smallest element and let $w = f(z) < z$, so $f(w) < f(z) = w$, so $w \in S$, contradiction.

Corollary. The only automorphism of a well-ordered set is the identity. Suppose $f : W \rightarrow W$ is an automorphism. $f(w) \geq w$ for all $w \in W$. $f^{-1}(w) \geq w$ for all $w \in W$. $w \geq f(w)$ for all $w \in W$. Finally, $w = f(w)$ for all $w \in W$.

If two well-ordered sets are isomorphic, then the isomorphism of W_1 onto W_2 is unique. Suppose $f, g : W_1 \rightarrow W_2$ are isomorphisms. Then $g^{-1} \circ f : W_1 \rightarrow W_1$ is an automorphism. Hence $g^{-1} \circ f \equiv \text{id}_{W_1}$. It follows $f \equiv g$.

Sps $f : W \rightarrow S$, where $S \subset W$ is an isomorphism. Then $S = W$. $f(w) \geq w$ for all $w \in W$. $f^{-1}(f(w)) \geq f(w)$ for all $w \in W$, so $f(w) = w$ for all $w \in W$. It follows $S = W$.

Let S_u denote the segment $\{w \in W : w < u\}$. If u is the smallest element of W , then $S_u = \emptyset$. Assume otherwise. Let $f : W \rightarrow S_u$ be an isomorphism. Consider the restriction $f|_{S_u} : S_u \rightarrow S_u$. From earlier, f is the identity id_{S_u} . Since f is injective, we have $f = f|_{S_u}$, so $S_u = W$. However, $u \in W$, contradiction.

Proof: If $\text{rng}(f) = \{w \in W : w < u\}$, then $f(u) < u$, contradiction.

Let $f = \{(x, y) \in W_1 \times W_2 : W_1(x) \cong W_2(y)\}$. Suppose $(x, y), (x, y') \in f$. Then $W_2(y) \cong W_2(y')$. If $y = y'$, then f is okay. Else WLOG $y < y'$, $y \in W_2(y')$ and $W_2(y)$ is a segment of $W_1(y')$, contradiction. Similarly, if $(x, y), (x', y) \in f$, then $x = x'$.

$f \neq \emptyset$. Let m, n be the smallest elements of W_1, W_2 , respectively. Then $W_1(m) = \emptyset = W_2(m)$.

Let $x \in W_1$, $y \in W_2$ and $h : W_1(x) \rightarrow W_2(y)$ be an isomorphism. Then $h(x) = y$. Suppose $h(x) < y$. Then $h(x) \in W_2(y)$ and $W_2(h(x)) \subset W_2(y)$, so $W_2(y)$ is isomorphic to a segment of itself, contradiction. Else $h(x) > y$. Apply the argument to $h^{-1}(y) < x$. It follows $h(x) = y$.

Suppose $x' < x$. We have $h(x') < h(x) = y$, so $h(x') \in W_2(y)$. Hence if $W_1(x') \subset W_1(x)$, then $W_1(h(x')) \subset W_2(y)$.

Suppose $a, b \in W$. Then $W(a) \subset W(b)$ iff $a \leq b$. If $a \leq b$, let $x \in W(a)$. Then $x < a$, so $x < a \leq b$, or $x < b$, or $x \in W(b)$. $W(a) \subset W(b)$.

Suppose $W_1(x) \cong W_2(y)$ and h is the isomorphism. Let $x' < x$. Then $W_1(x') \cong W_2(h(x'))$. Let $u \in W_1(x')$. Then $h(u) < h(x')$, so $h(u) \in W_2(h(x'))$. $h : W_1(x') \rightarrow W_2(h(x'))$ is a well-defined map. Suppose $v \in W_2(h(x'))$. Then $v < h(x')$, so $h^{-1}(v) < x'$, or $h^{-1}(v) \in W_1(x')$. h is surjective. h is injective and increasing by givens, so h is indeed an isomorphism.

If $x' < x$, then $W_1(x') \cong W_2(h(x')) \subset W_2(y)$.

If $\text{dom}(f) = W_1$ and $\text{rng}(f) = W_2$, then $W_1 \cong W_2$. Pick any $w \in W_1$. Pick???????

For well ordered sets W_1 and W_2 , show that exactly one of the following is true:

- (i) W_1 is isomorphic to W_2
- (ii) W_1 is isomorphic to a segment of W_2

(iii) W_2 is isomorphic to a segment of W_1

Let W_1 and W_2 be well ordered sets. Suppose for all $w_1 \in W_1$ there exists $w_2 \in W_2$ such that $W_1(w_1) \cong W_2(w_2)$ and conversely for all $w_2 \in W_2$, there exists a corresponding w_1 . Show that $W_1 \cong W_2$. Suppose W_1 has a largest element m_1 . Then $W_1(m_1) \cong W_2(w_2)$.

Suppose W_1 has a largest element m_1 . Then $W_1(m_1) \cong W_2(w_2)$ for some w_2 . Suppose FTSOC there exists an element in W_2 greater than w_2 , say m_2 . Then $W_2(m_2) \cong W_1(m_1)$, but the isomorphism is no longer bijective, contradiction. Hence, m_2 is the greatest element of W_2 and $W_1(m_1) \cup \{m_1\} \cong W_2(m_2) \cup \{m_2\}$.

Otherwise, let $w_1 \in W_1$ be arbitrary. There exists $w'_1 > w_1$. Consider $W_1(w'_1) \cong W_2(w'_2)$ for some $w'_2 \in W_2$. Define $f(w_1) = h(w_1)$. $f(w_1)$ is well-defined because two isomorphisms covering w_1 descend to the same isomorphism over their common domain. Hence, we have $f : W_1 \rightarrow W_2$. f is order preserving and bijective, so $W_1 \cong W_2$.

Suppose $\text{dom}(f) \neq W_1$. Let m_1 be the smallest element of $W_1 - \text{dom}(f)$. $\text{dom}(f) = W_1(m_1)$. Suppose also $\text{rng}(f) \neq W_2$, so $\text{rng}(f) = W_2(m_2)$. Then $W_1(m_1) \cong W_2(m_2)$, so $m_1 \in \text{dom}(f)$, a contradiction.

Suppose $\text{dom}(f) = W_1$ and $\text{rng}(f) \neq W_2$. Write $\text{rng}(f) = W_2(m_2)$. Any segment of W_1 is isomorphic to some segment of $W_2(m_2)$, and vice versa. Hence, $W_1 \cong W_2(m_2)$.

Define $f = \{(x, y) \in W_1 \times W_2 : W_1(x) \cong W_2(y)\}$.

Prop 1. Well ordered set isomorphism is an equivalence relation.

Let W be a well ordered set. Then id_W will do for $W \cong W$. Suppose $f : A \rightarrow B$ is an isomorphism. Then $f^{-1} : B \rightarrow A$ is bi-increasing and bijective, so $B \cong A$. Suppose $f : A \rightarrow B$ and $g : B \rightarrow C$ are isoms. Then $g \circ f : A \rightarrow C$ is bijective and bi-increasing.

Lemma 0: Let W be a well ordered set such that $W(x) \cong W(y)$. Then $x = y$.

Lemma 1: f is a well-defined function and is injective.

Lemma 2: If $x \in \text{dom}(f)$, then $W_1(x) \subset \text{dom}(f)$. Similarly, if $y \in \text{dom}(f)$, then $W_2(y) \subset \text{rng}(f)$.

Lemma 3: If W_1, W_2 are well ordered sets such that every initial segment of W_1 is isomorphic to an initial segment of W_2 and vice versa, then $W_1 \cong W_2$.

PL 0: Suppose $x < y$. Then $W(x) \subset W(y)$, contradiction, as $W(y)$ is a woset.

$x > y$ also gives a cont. It follows $x = y$.

PL 1: Suppose $(x, y), (x, y') \in f$. Then $W(y) \cong W(x)$ and $W(y') \cong W(x)$, so $W(y) \cong W(y')$, so $y = y'$ by L0. f is well-defined. Similarly, if $(x', y), (x, y) \in f$ gives $x = x'$, so f is injective.

PL 2: Suppose $x \in \text{dom}(f)$. There exists $y \in W_2$ such that $h : W_1(x) \cong W_2(y)$. Note that $W_1(x') \cong W_2(h(x'))$. Hence, $x' \in \text{dom}(f)$, or that $W_1(x) \subset \text{dom}(f)$. Similarly, if $y \in \text{rng}(f)$, then $W_2(y) \subset \text{rng}(f)$.

PL 3: Let I_1 denote the set of initial segments of W_1 . Define I_2 analogously. There exists a function $f : I_1 \rightarrow I_2$. Suppose $W_1(x_1) \cong W_2(x_2)$ and $W_1(x'_1) \cong W_2(x_2)$. Then $W_1(x_1) \cong W_1(x'_1)$, so $x_1 = x'_1$. Hence f is injective. There exist injective functions $f : I_1 \rightarrow I_2$ and $g : I_2 \rightarrow I_1$. Then there exists bijective $h : I_1 \rightarrow I_2$?

If $\text{dom}(f) = W_1$ and $\text{rng}(f) = W_2$, then $W_1 \cong W_2$. If $\text{dom}(f) = W_1$ and $\text{rng}(f) \neq W_2$, then write $\text{rng}(f) = W_2(m_2)$. Hence $W_2(m_2) \cong W_1$. Else $W_1(m_1) \cong W_2(m_2)$, so $(m_1, m_2) \in f$, contradiction.

1.9. Show that every $n \in N$ is T-finite. Let $A = \{n \in N : n \text{ is T-finite}\}$. $\emptyset \in A$. Suppose $a \in A$. Consider $a \cup \{a\}$. Let $X \subset P(a \cup \{a\})$. X reduces to a set $X' = \{x \in P(a) : x \cup \{a\} \in X\}$. If $X' = \emptyset$, then $X \subset \{\emptyset, \{a\}\}$, which works. Else let b be a maximal element of X' . If $y = b \cup \{a\} \in X$, then y is maximal in X . If $y = b \cup \{a\} \notin X$, then b is maximal. Hence, $a \cup \{a\}$ is T-finite. A is inductive and a subset of N , so $A = N$.

If X is transitive, then so is $P(X)$. Let $Y \in P(X)$, or that $Y \subset X$. For all $y \in Y$, $y \in X$, so $y \subset X$. Hence, $Y \subset P(X)$. N is transitive, so $N \subset P(N)$.

For any $n \in N$ and $n \neq 0$, there exists $m \in N$ such that $n = m \cup \{m\}$. Then m is unique. Suppose $m \cup \{m\} = m' \cup \{m'\}$. If $m = m'$, we're done. Else assume $m \in m'$ and $m' \in m$. Each element of N is transitive, so $m \subset m'$ and $m' \subset m$, so $m = m'$. Hence, for $n \neq 0$ we may write $n - 1$ to denote m .

1.11. N is T-infinite. Consider $N \subset P(N)$. For any $n \in N$, $n \subsetneq (n + 1)$ and $n + 1 \in N$ as well.

If $\alpha \neq \beta$ are ordinals and $\alpha \subset \beta$, then $\alpha \in \beta$. Let $a \in \alpha$ and $b < a$. Then $b \in a \subset \alpha$, or $b \in \alpha$. α is an initial segment of β . Write $\alpha = \beta(\gamma)$ for some $\gamma \in \beta$. This is trivially $\alpha = \beta$, so $\alpha \in \beta$.

Consider $\gamma = \alpha \cap \beta$. Suppose FTSOC $\gamma \neq \alpha$ and $\gamma \neq \beta$. Then $\gamma \in \alpha \cap \beta$, contradiction. Hence, either $\alpha \subset \beta$ or $\beta \subset \alpha$.

Let S be a set with the least upper bound property. Let $\mathcal{A}$ be a collection

of sets $A \subset S$. Is it true $\sup_{A \in \mathcal{A}} \sup(A) = \sup \bigcup \mathcal{A}$?

Is it true for ordinals $\sup(d + A) = d + \sup(A)$?